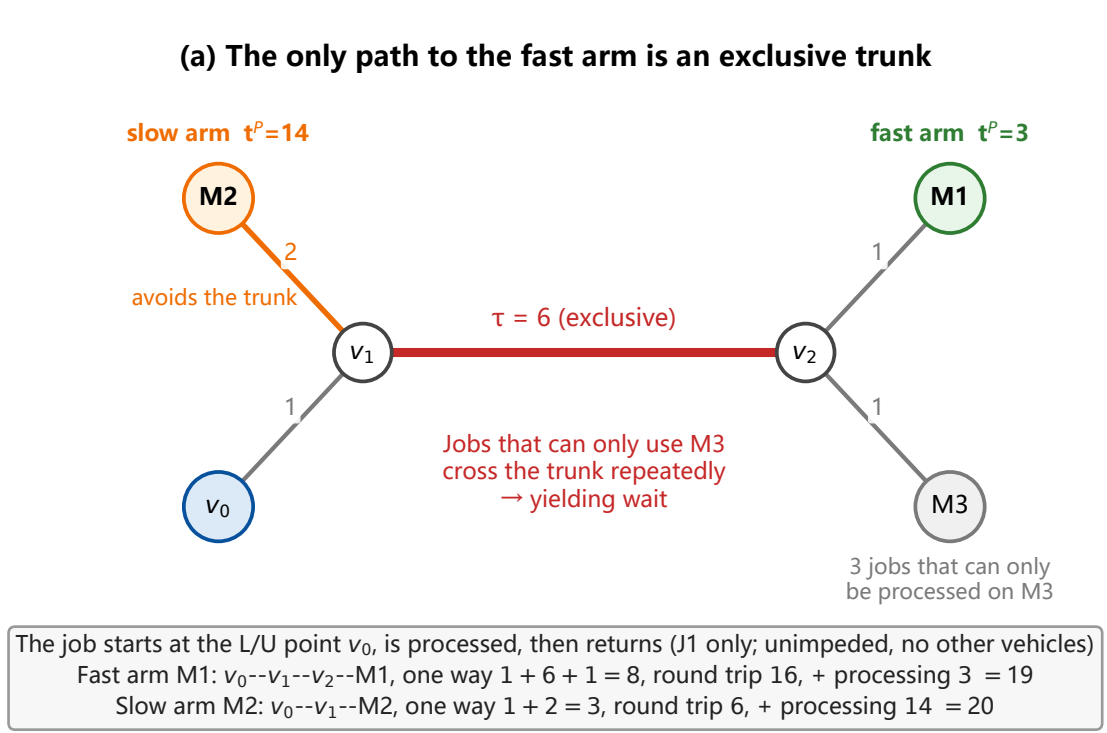
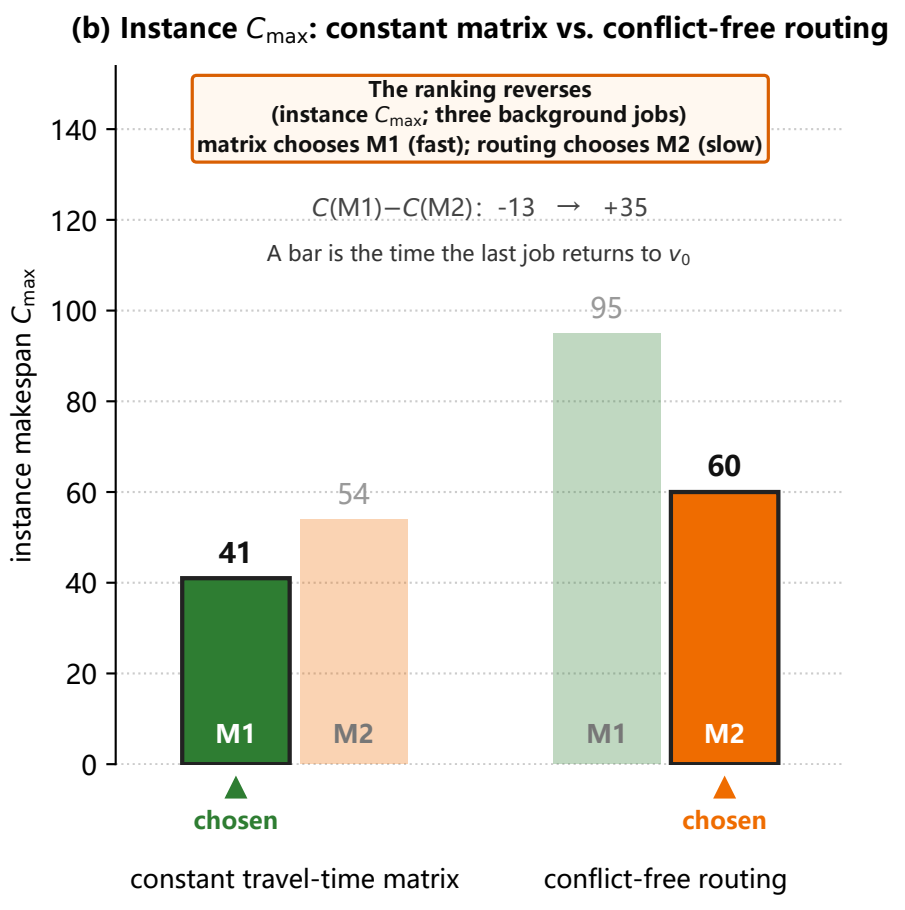

(a) The only path to the fast arm is an exclusive trunk
slow arm t^P=14
fast arm t^P=3
M2
M1
2
1
avoids the trunk
τ = 6 (exclusive)
v_1
v_2
1
1
Jobs that can only use M3
cross the trunk repeatedly
→ yielding wait
v_0
M3
3 jobs that can only
be processed on M3
The job starts at the L/U point v_0, is processed, then returns (J1 only; unimpeded, no other vehicles)
Fast arm M1: v_0--v_1--v_2--M1, one way 1 + 6 + 1 = 8, round trip 16, + processing 3 = 19
Slow arm M2: v_0--v_1--M2, one way 1 + 2 = 3, round trip 6, + processing 14 = 20


(b) Instance C_{max}: constant matrix vs. conflict-free routing
The ranking reverses
(instance C_{max}: three background jobs)
matrix chooses M1 (fast); routing chooses M2 (slow)
C(M1)−C(M2): -13 → +35
A bar is the time the last job returns to v_0
instance makespan C_{max}
0
20
40
60
80
100
120
140
41
54
95
60
M1
M2
M1
M2
chosen
chosen
constant travel-time matrix
conflict-free routing